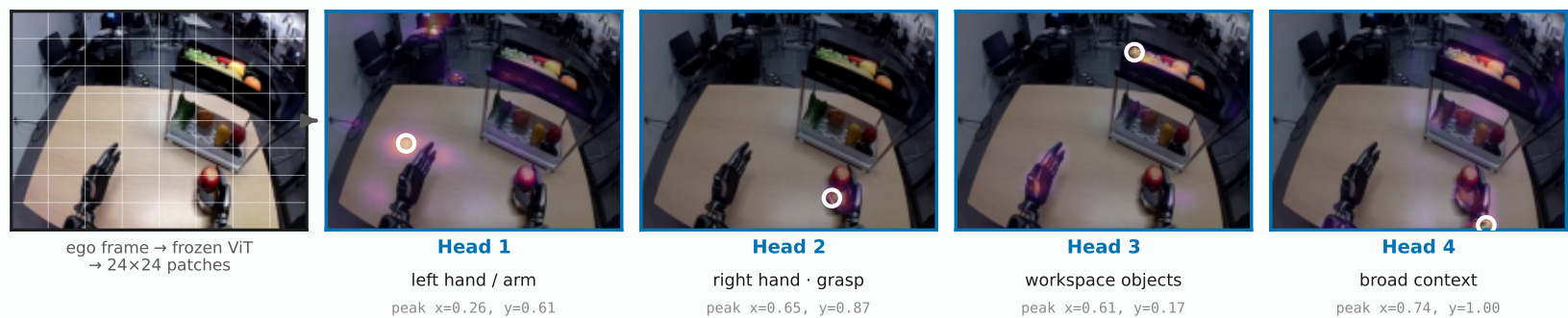


KineScore

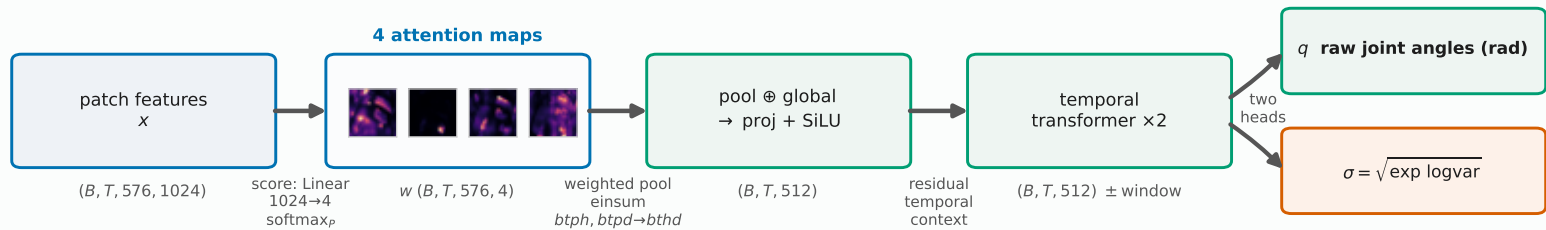
an attentive head reads the joint angles from pixels, then every physics score is a geometric measurement on the robot's own URDF model

PART A Reading the pose from pixels — the attentive pose head

A1 the 4 pooling heads $w = \text{softmax}_p(\text{score}(x))$ specialize — each head's softmax mass over the 24x24 patch grid, on one real frame



A2 the data flow: how patch features become a pose (tensor shapes B batch · T frames · P=576 patches · D=1024)



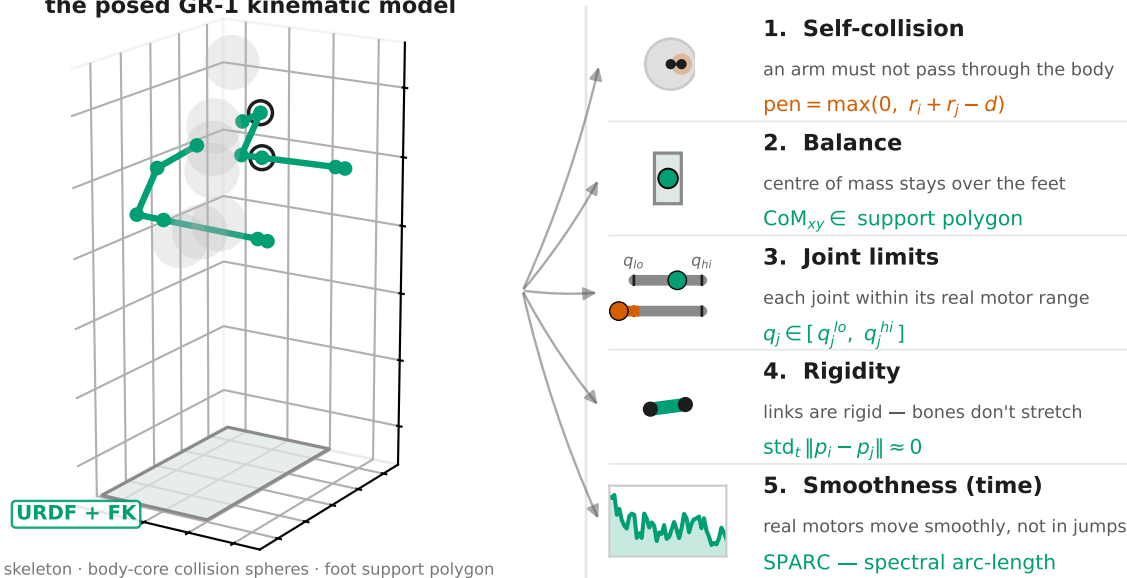
A3 grounding — per-joint gradient saliency: each joint's angle is read from its OWN limb, not the whole image



place the joint angles q on the robot's own CAD model (URDF + forward kinematics)

PART B Checking the physics — geometrically, on that one URDF model

the posed GR-1 kinematic model



Distribution-free: every score is a geometric measurement on the robot's own model— so real footage can never be ranked below generated.